Mid-Semester Test
(EC-2 Regular)

Course No.	: SS ZG526
Course Title	: DISTRIBUTING COMPUTING
Nature of Exam	: Closed Book
Weightage	: 30%
Duration	: 2 Hours
Date of Exam	: 23/09/2017 (AN)

No. of Pages = 2
No. of Questions = 5

Note:
1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1 (a) Discuss the 4 different features of distributed systems. [4]

Q.1 (b) Describe the causal ordering model of communication. [1]

Q.1 (c) Define cut of a distributed computation. [1]

Q.2 (a) Consider the space-time diagram shown in Fig. 1 depicting 3 processes.

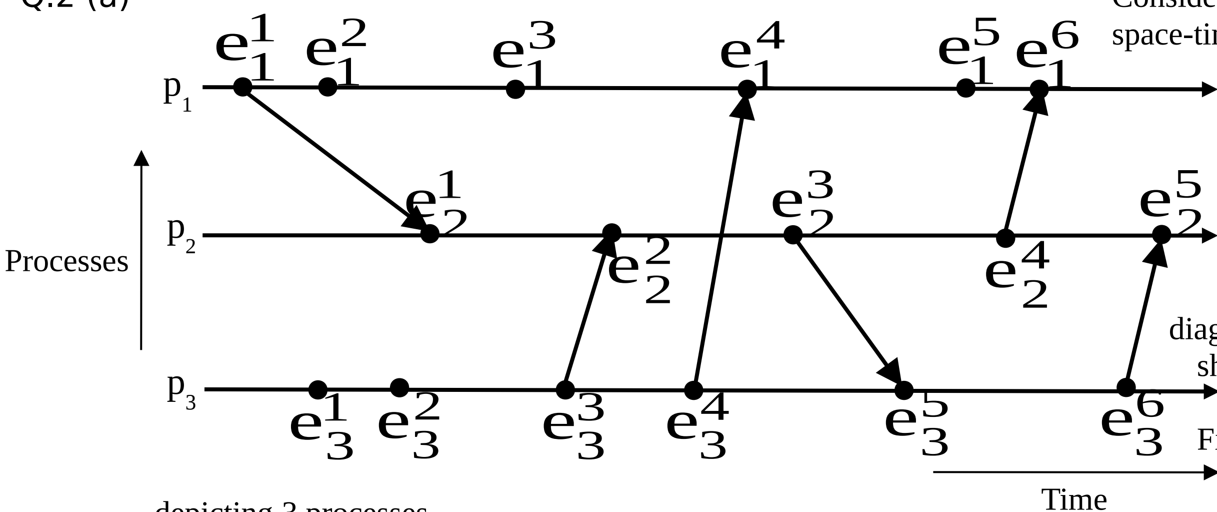

Fig. 1

Are the following pairs of events causally related? Justify your answer. [1 x 4 = 4]

(i) e_1^1 and e_3^5 (ii) e_3^1 and e_2^5 (iii) e_1^3 and e_3^6 (iv) e_2^2 and e_1^6

Q.2 (b) For the space-time diagram of Fig. 1, is the global state $\{LS_1^2, LS_2^2, LS_3^4\}$ consistent? Give reasons in support of your answer. [1]

Q.3 (a) Identify the scalar clock values of the events and the timestamps of the messages exchanged among the 3 processes shown in Fig. 1. Initially, the scalar clocks of all the 3 processes are 0. Use **R1** and **R2** for updating the scalar clocks. Assume *d* = 1. [4]

Q.3 (b) What are the heights of events e_3^5 and e_1^6 in Fig. 1? [½ + ½ = 1]

Q.3 (c) Is the expression $(e_x \parallel e_y) \wedge (e_y \parallel e_z) \Rightarrow e_x \parallel e_z$ correct for any 3 events e_x, e_y and e_z? Explain your answer. [1]

Q.4 (a) Write a short note on Butterfly network by clearly describing the interconnection function and the routing function. Note that your answer should be accompanied with a diagram illustrating the interconnection pattern for an 8-input and 8-output Butterfly network. [2 + 2 = 4]

Q.4 (b) What are the 2 conditions that must be satisfied by a global state so that we can say that it is a consistent global state? [2]

Q.5 (a) Determine a BFS spanning tree by applying the synchronous single-initiator spanning tree algorithm using flooding on the graph shown in Fig. 2. Assume node A to be the root node. Clearly show the round numbers in which the QUERY messages are sent by the different nodes. [5]

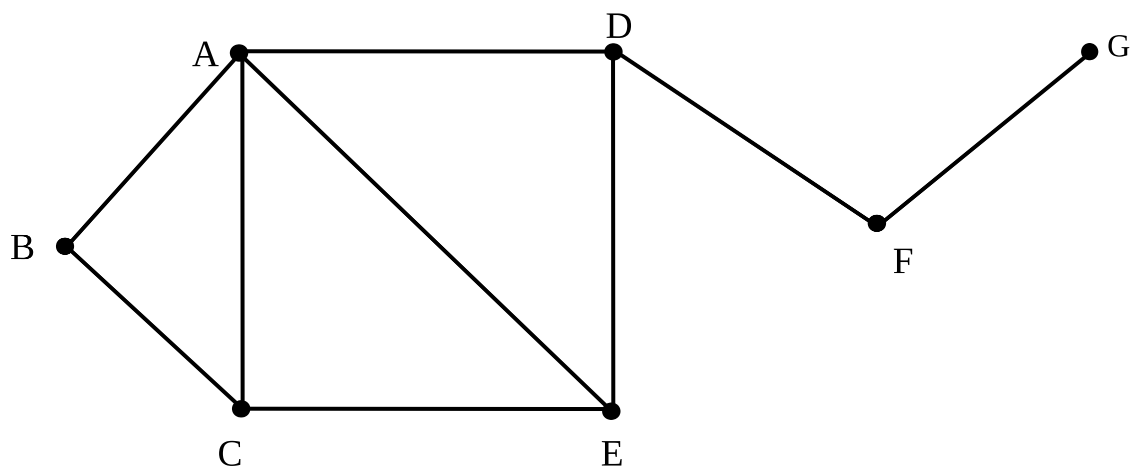

Fig. 2

Q.5 (b) Describe the different steps of the Birman-Schiper-Stephenson Protocol. [2]

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